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Constant Acceleration 2i

Subject & topics: Maths **Stage & difficulty:** A Level P1

A stone is released from rest on a bridge and falls vertically into a lake. The stone has velocity $14\,\mathrm{m\,s^{-1}}$ when it enters the lake.

Part A

Distance fallen

Calculate the distance the stone falls before it enters the lake.

Part B

Time on entering the lake

Calculate the time after its release when it enters the lake. Give your answer to 3 significant figures.

Part C

Acceleration

The lake is $15\,\mathrm{m}$ deep and the stone has velocity $20\,\mathrm{m\,s^{-1}}$ immediately before it reaches the bed of the lake.

Given that there is no sudden change in the velocity of the stone when it enters the lake, find the acceleration of the stone while it is falling through the lake. Give your answer to 2 significant figures.

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Vectors: Diagrams and Proof 1i

Subject & topics: Maths **Stage & difficulty:** A Level P2

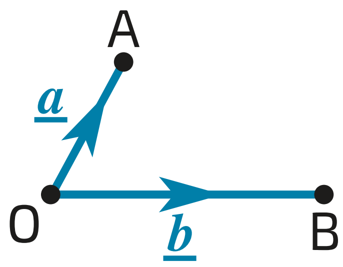


Figure 1: Three points O, A and B, and vectors joining them.

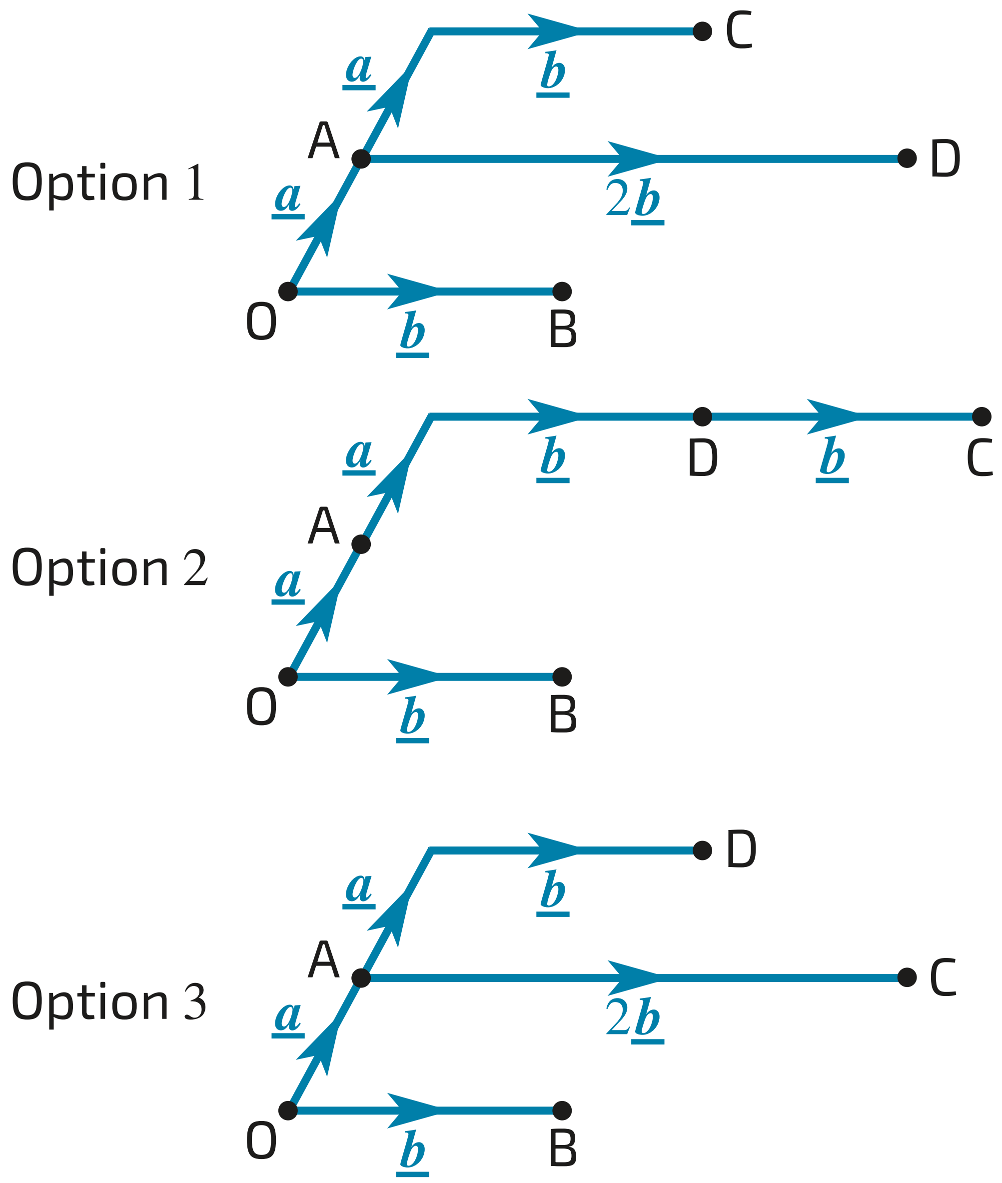
Figure 1 shows points O, A and B, with $\overrightarrow{\mathrm{OA}} = \underline{\boldsymbol{a}}$ and $\overrightarrow{\mathrm{OB}} = \underline{\boldsymbol{b}}$.

Part A

Sketch

Make a sketch of the diagram and mark the points C and D such that $\overrightarrow{OC} = \underline{\boldsymbol{a}} + 2\underline{\boldsymbol{b}}$ and $\overrightarrow{OD} = 2\underline{\boldsymbol{a}} + \underline{\boldsymbol{b}}$.

Choose the correct sketch from the 3 options below.



- Option 1
- Option 2
- Option 3

Part B

Vector $\overrightarrow{DC}$

Express $\overrightarrow{DC}$ in terms of $\underline{\boldsymbol{a}}$ and $\underline{\boldsymbol{b}}$, simplifying your answer.

$$\overrightarrow{DC} = \square\ \underline{\boldsymbol{a}} + \square\ \underline{\boldsymbol{b}}$$

Part C

Proof

Prove that ABCD is a parallelogram.

Sides $\overrightarrow{AD}$ and $\square$ are both equal to $\square$. Therefore they are parallel and of equal length.

Sides $\overrightarrow{AB}$ and $\square$ are both equal to $\square$. Therefore they are parallel and of equal length.

The quadrilateral ABCD has two pairs of parallel sides of equal length. Therefore ABCD is a parallelogram.

Items:

$\underline{\boldsymbol{b}} - \underline{\boldsymbol{a}}$ $\overrightarrow{BC}$ $\overrightarrow{CD}$ $\overrightarrow{DC}$ $\overrightarrow{CB}$ $\underline{\boldsymbol{a}} - \underline{\boldsymbol{b}}$ $\underline{\boldsymbol{a}} + \underline{\boldsymbol{b}}$ $-\underline{\boldsymbol{a}} - \underline{\boldsymbol{b}}$

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Kinematics Graphs 3i

Subject & topics: Maths **Stage & difficulty:** A Level P1

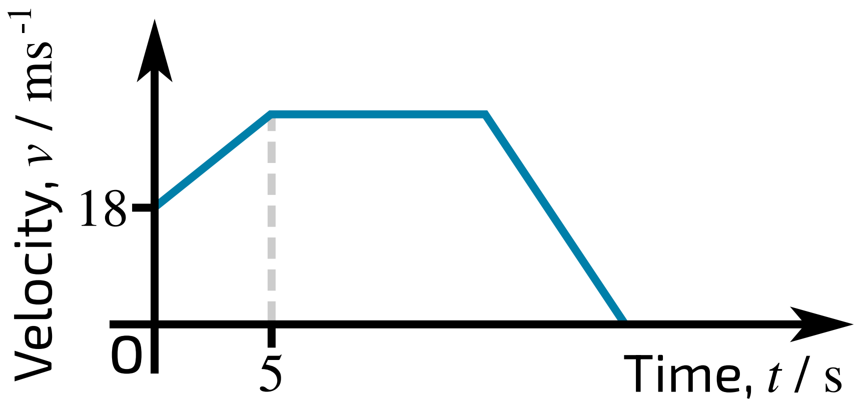

Figure 1: (t, v) graph of a car moving along a straight road.

Figure 1 shows the (t, v) graph of a car moving along a straight road, where $v\,\mathrm{m\,s^{-1}}$ is the velocity of the car at time $t\,\mathrm{s}$ after it passes through the point A. The car passes through A with velocity $18\,\mathrm{m\,s^{-1}}$, and moves with constant acceleration $2.4\,\mathrm{m\,s^{-2}}$ until $t = 5$. The car subsequently moves with constant velocity until it is $300\,\mathrm{m}$ from A. When the car is more than $300\,\mathrm{m}$ from A, it has constant deceleration $6\,\mathrm{m\,s^{-2}}$, until it comes to rest.

Part A
Greatest speed

Find the greatest speed of the car.

Part B
Deceleration

Find the value of t for the instant when the car begins to decelerate.

Part C

Distance from A

Calculate the distance from A of the car when it is at rest.

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isaac You work it out.

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Simple Force Diagrams 1ii

Subject & topics: Maths **Stage & difficulty:** A Level P1

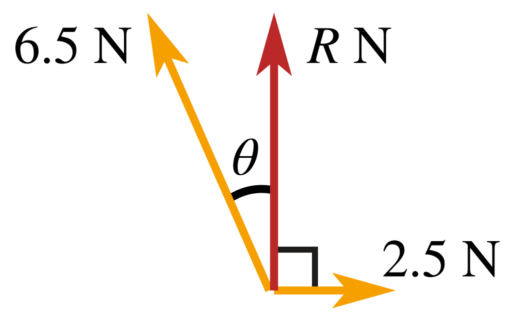

Figure 1: Diagram of 3 forces acting at a point.

Forces of magnitudes $6.5\,\mathrm{N}$ and $2.5\,\mathrm{N}$ act at a point in the directions shown in **Figure 1**. The resultant of the two forces has magnitude $R\,\mathrm{N}$ and acts at right angles to the force of magnitude $2.5\,\mathrm{N}$.

Part A

Finding θ

Find the angle θ correct to 3 significant figures.

Part B

Finding R

Find the value of R.

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isaac You work it out.

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Trigonometry: Identities and Equations 3ii

Subject & topics: Maths **Stage & difficulty:** A Level P1

Solve each of the following equations for $0^\circ \leq x \leq 180^\circ$.

Part A

Equation 1

Solve the following equation for $0^\circ \leq x \leq 180^\circ$:

$$\sin 2x = 0.5$$

$^\circ$

Part B

Equation 2

Solve the following equation for $0^\circ \leq x \leq 180^\circ$:

$$2\sin^2 x = 2 - \sqrt{3}\cos x$$

$^\circ$

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Sine and Cosine Rules and Area 1ii

Subject & topics: Maths **Stage & difficulty:** A Level P1

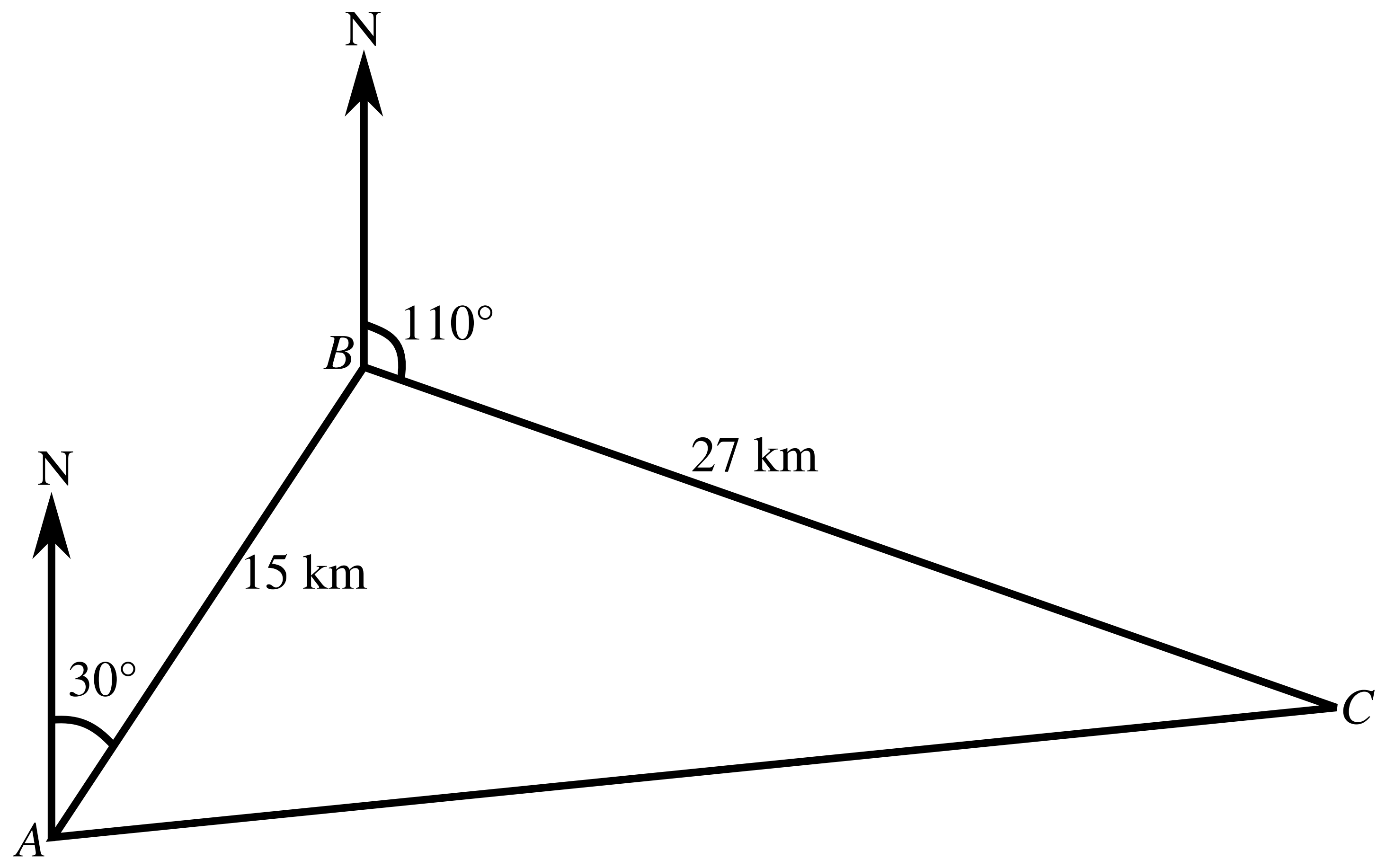

Figure 1: The journey of a ilfeboat.

In **Figure 1**, a lifeboat station is at point A. A distress call is received and the lifeboat travels $15\,\mathrm{km}$ on a bearing of $030\,^{\circ}$ to point B. A second call is received and the lifeboat then travels $27\,\mathrm{km}$ on a bearing of $110\,^{\circ}$ to arrive at point C. The lifeboat then travels back to the station at A.

Part A

Angle ABC

Find the value of angle ABC in degrees to three significant figures.

Part B

Distance

Find the distance the lifeboat has to travel to get from C back to A. Give your answer in kilometres to three significant figures.

Part C

Bearing

Find the bearing on which the lifeboat has to travel to get from C to A. Give your answer in degrees to three significant figures.

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Another River Crossing

Subject & topics: Maths | Geometry | Vectors **Stage & difficulty:** A Level P1

A rowing boat sets out to cross a river. The banks of the river are parallel, the width of the river is $80\,\mathrm{m}$, and the river flows downstream at $2\,\mathrm{ms}^{-1}$. The boat is directed at an angle of 60° to the bank, and the rowers start rowing at a speed u relative to the water. This is illustrated in **Figure 1**.

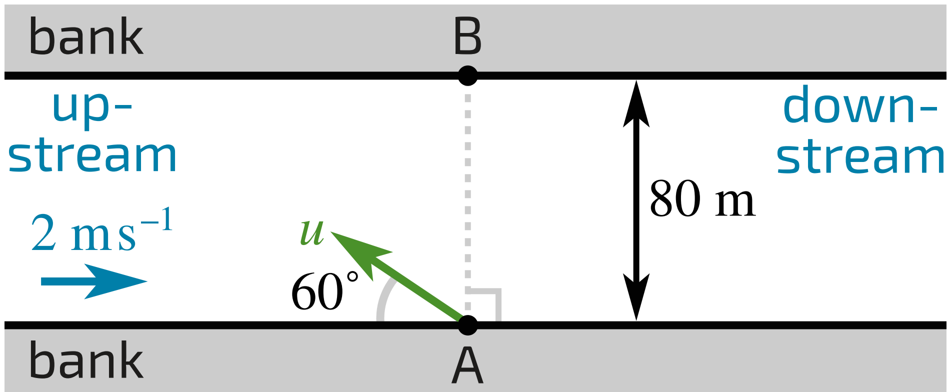

Figure 1: Illustrating the rowers setting off on their crossing.

In **Figure 1** the point at which the rowers leave the bank is A. Point B is on the opposite bank and lies on a line perpendicular to the banks through A.

The rowers arrive at a point on the opposite bank that is $20\,\mathrm{m}$ upstream from B. Assuming that the rowers maintain both their angle to the starting bank and their speed throughout the crossing, find the speed u. Give your answer to 2 sf.

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